

ELECTRONICS LABORATORY SIMULATION REPORT

Number of the Experiment : 4

Students : Emirhan Ak - 151220194038, Mustafa Yağcı - 151220192027, Emre Saraç - 151220192091

Date : 18/11/2022

Group : 1B

EXPERIMENT NO: 4 Common Emitter BJT Amplifiers

1. Build the circuit as in Figure 1. Set $V_{CC}=12V$, $R_1=10\text{ Kohm}$, $R_2=2.7\text{ Kohm}$, $R_C=4.7\text{ Kohm}$, $R_L=R_E=2.7\text{ Kohm}$, $R_S=10k$, $C_{C1}=C_E=10\text{ }\mu F$ and $C_{C2}=1\text{ }\mu F$. Do not apply V_s from the signal generator. Measure V_{CEQ} , V_{BEQ} and I_{CQ} values. Determine the operation mode of BJT. Then calculate the I_{BQ} , g_m , r_{π} , r_e values of the transistor. (Assume that $\beta=250$, $V_T=0.025\text{ V}$).
2. Apply 0.1 Vpp 5 KHz sinusoidal signal to V_s . With the oscilloscope, monitor V_s with channel 1 and V_o with channel 2. Add V_s and V_o plottings to your report. Is there any phase difference between V_s and V_o ?
3. Comment on the effect of the C_{C1} and C_{C2} capacitors.
4. Remove C_e capacitor from the circuit. With the oscilloscope, monitor V_s with channel 1 and V_o with channel 2. Add V_s and V_o plottings to your report. Comment on the effect of the C_e capacitor. After that place C_e capacitor back.
5. Remove R_L from the circuit. With the oscilloscope, monitor V_s with channel 1 and V_o with channel 2. Add V_s and V_o plottings to your report. After that place R_L resistor back.
6. Change the V_s frequency step by step using the values given below. For each frequency, calculate the gain of the amplifier by using the measurements of V_{out} and V_s voltages. Plot Gain-Frequency graphic in your report.
Freq: 50 Hz-100 Hz-250 Hz-500 Hz-1 KHz-50 KHz-125 KHz-250 KHz-500 KHz-1 MHz
7. Calculate the input resistance, output resistance and voltage gain of the amplifier (Assume that $r_0 \gg R_L$ and $r_0 \gg R_C$). Compare the calculated and the measured gain values.

ELECTRONICS LABORATORY EXPERIMENT REPORT

Note: Your report should contain the followings: objectives, experimental studies and results, figures and conclusion. **In this part, you will use your lab environment outputs, and add your comments about the results.**

Objectives

- To learn how BJTs work when AC voltage is applied.
- How does capacitors work with BJTs.
- Determining the change in output voltage when the frequency of the AC source is changed.

Experimental Studies and Results

1) In the first part of the experiment, we simulated the circuit in Figure 1 on Falstad and compared the values we obtained in the real lab.

	Measured Values	Simulation Values
I_B	3.6 μ A	2.9 μ A
g_m	0.036	0.0288
r_π	6945	8620.9
r_e	27.77	34.48

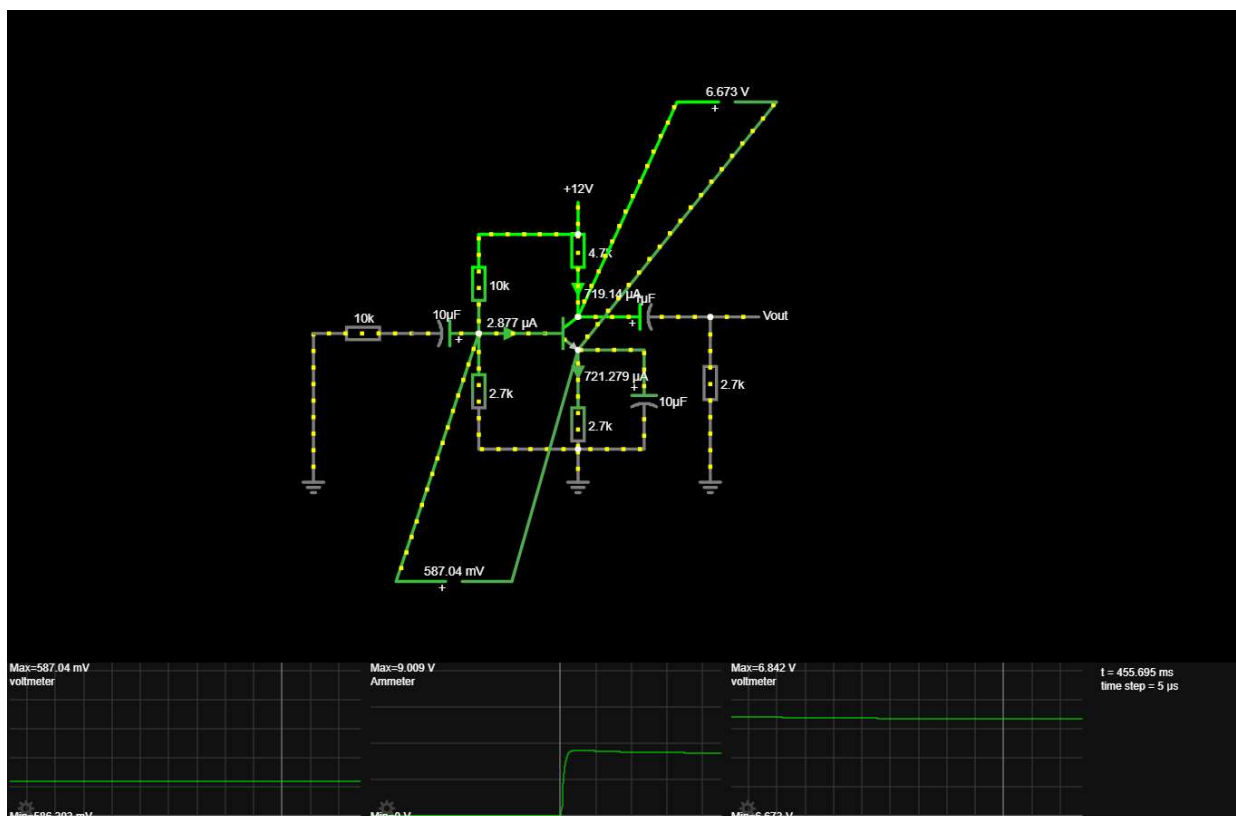


Figure 1

2) In the second part, we applied a sinusoidal voltage source to the circuit with a peak-to-peak voltage of 0.1V with 5kHz and scoped the V_s and V_o values. (We couldn't add the real values we calculated on the oscilloscope since the measurements on it were incorrect. Mr. Ahmet said we can remind him of this in our report.)

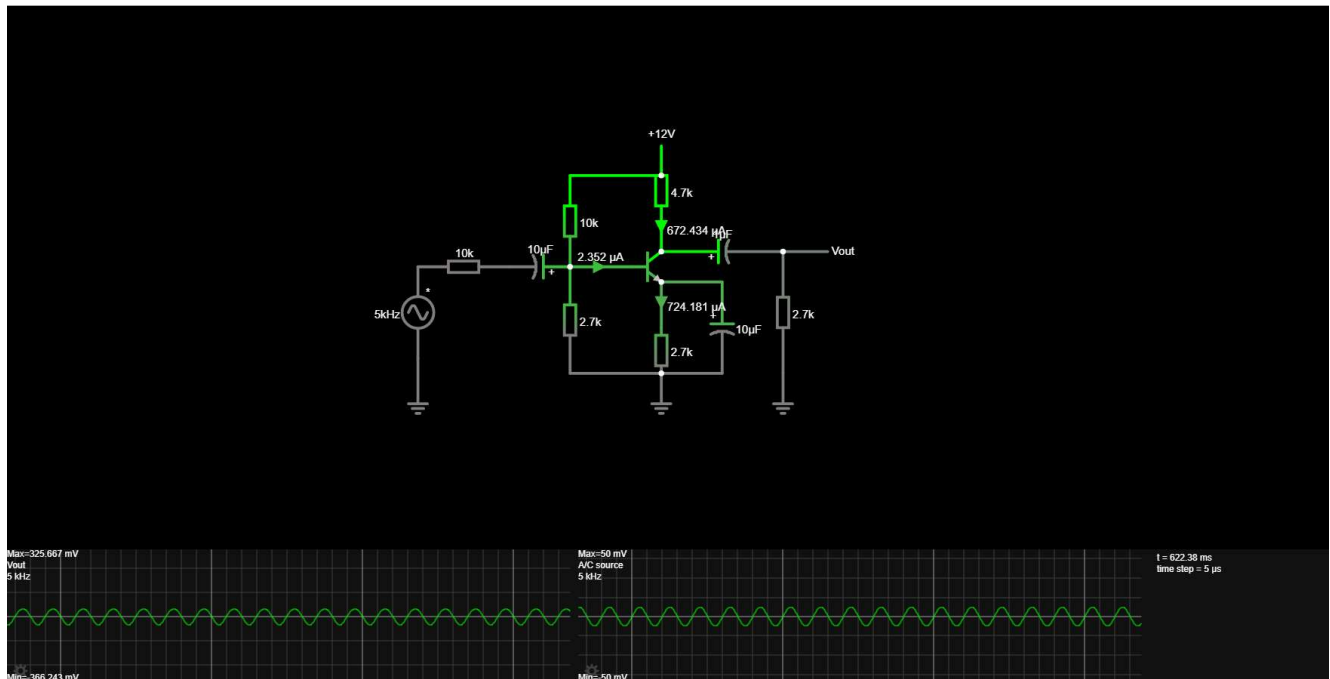


Figure 2

3) The capacitors C_{C1} and C_{C2} help us to convert the AC signal into DC signal.

4)

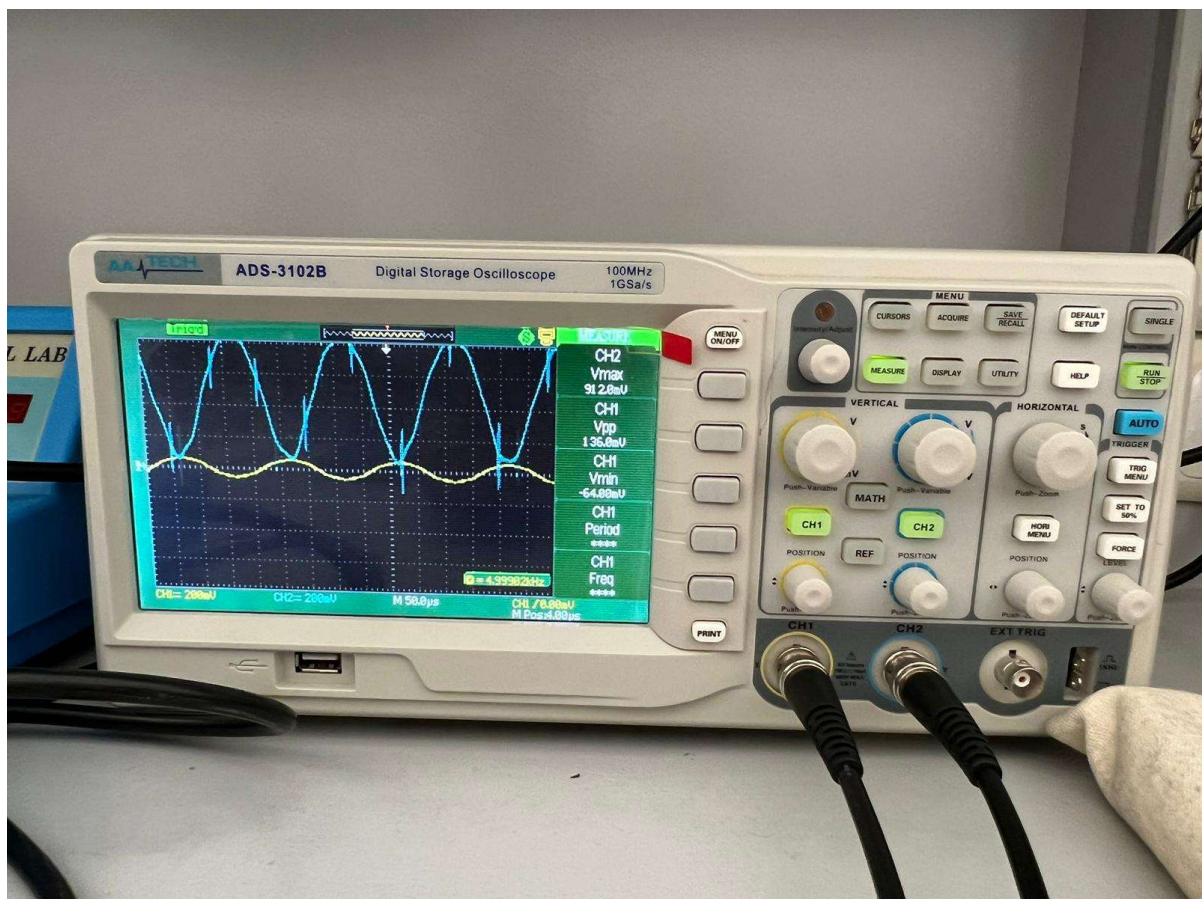


Figure 3

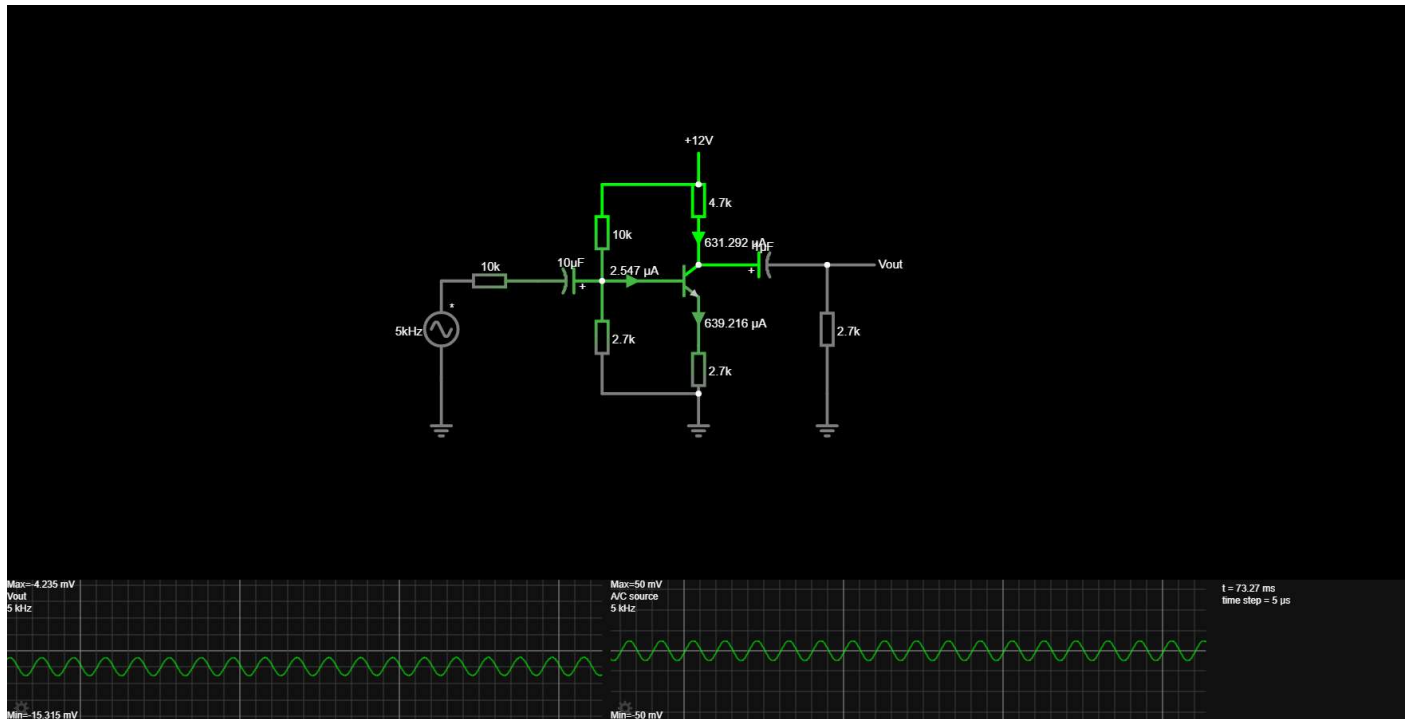


Figure 4

As you can see, the simulation and the oscilloscope measurements are quite different. This is because of the reason I've stated before.

5)

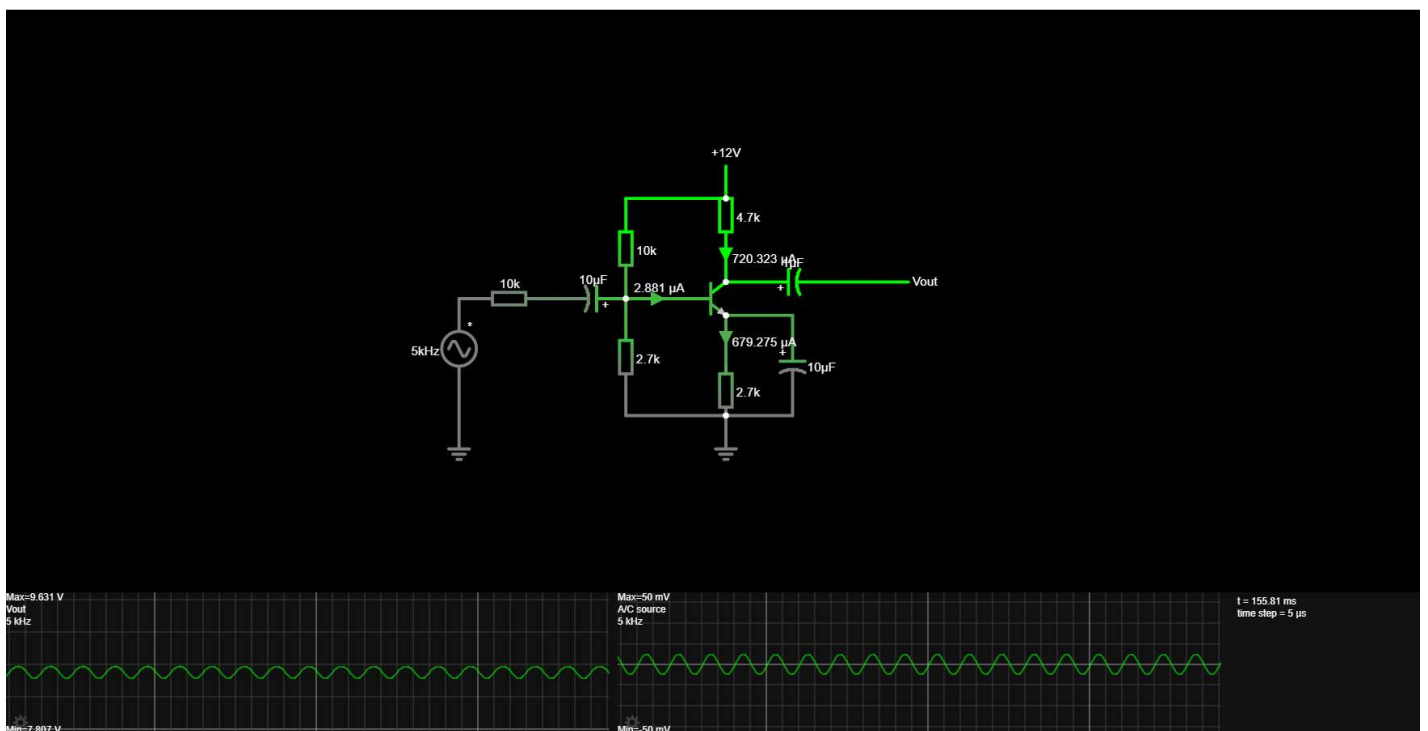


Figure 5

6)

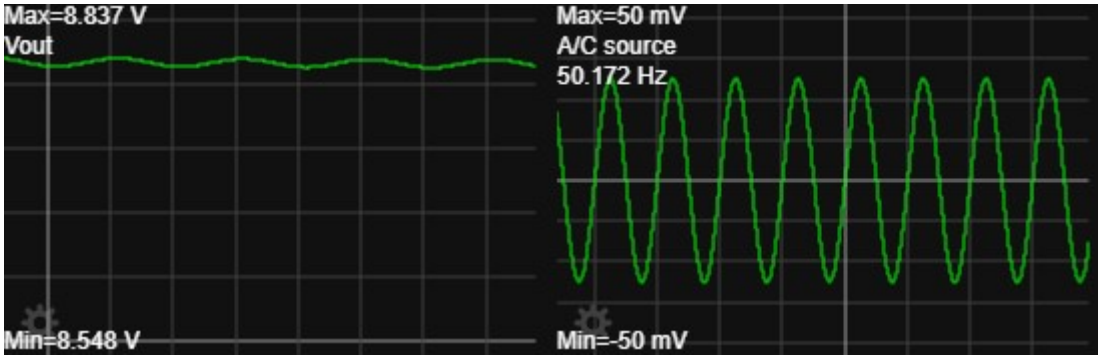


Figure 6 - 50 Hz

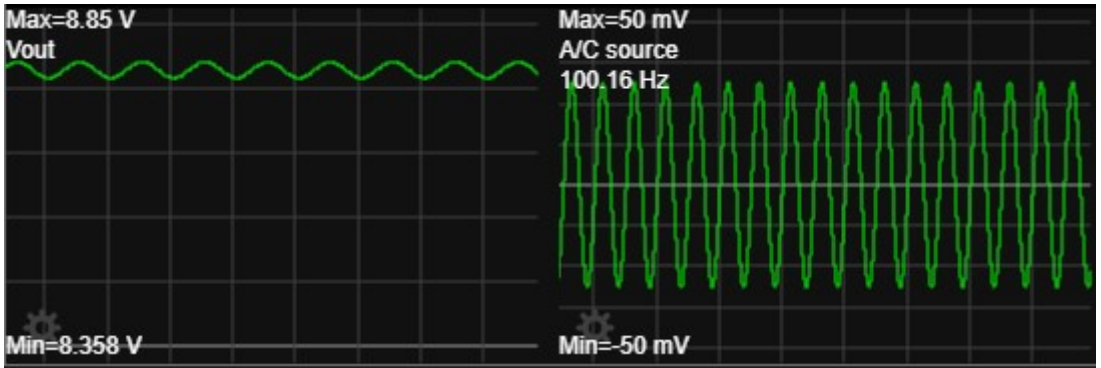


Figure 7 - 100 Hz

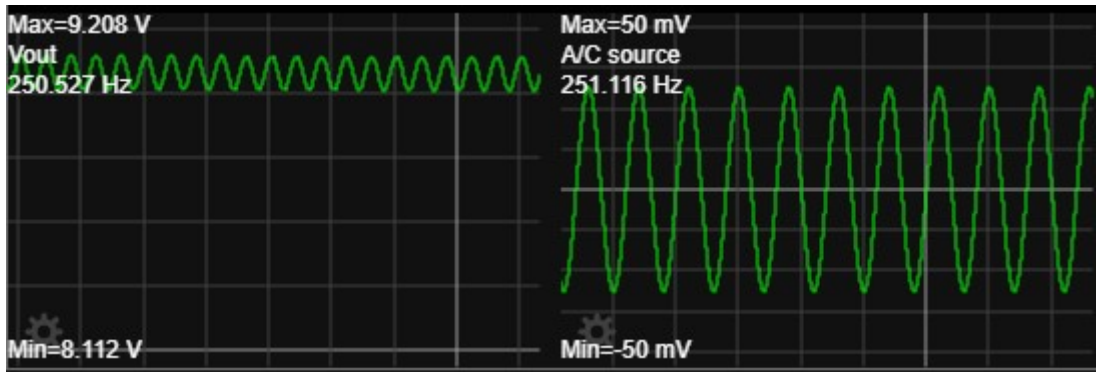


Figure 8 - 250 Hz

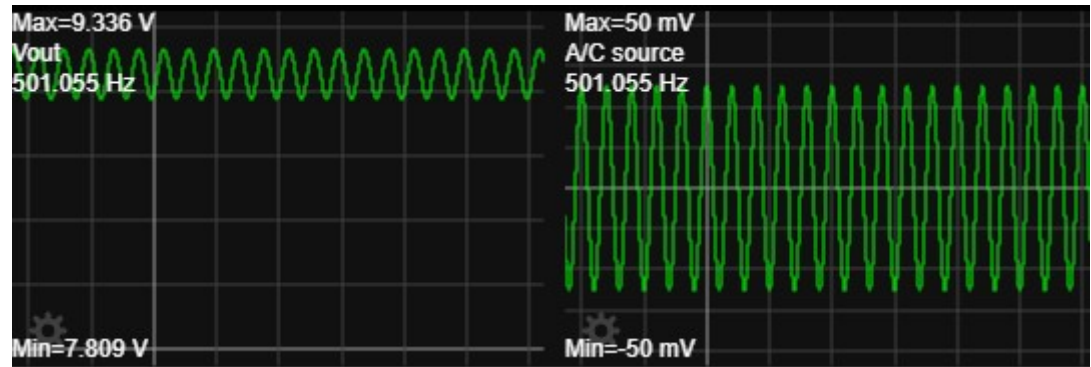


Figure 9 - 500 Hz

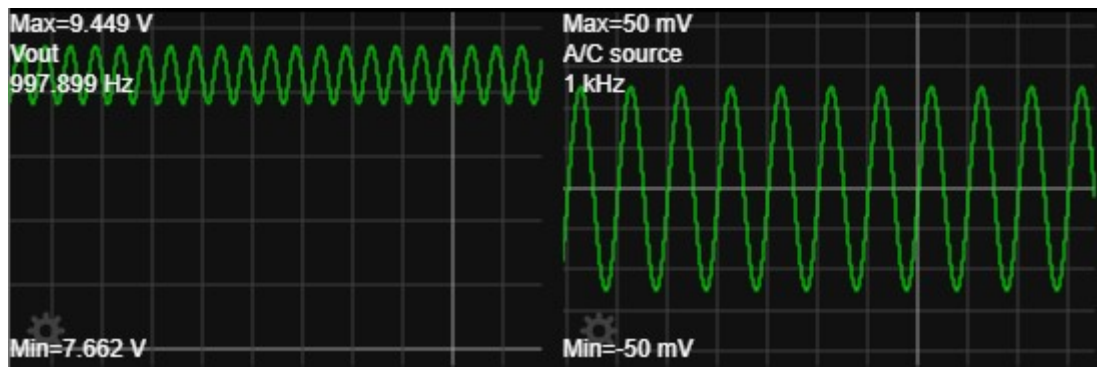


Figure 10 - 1kHz

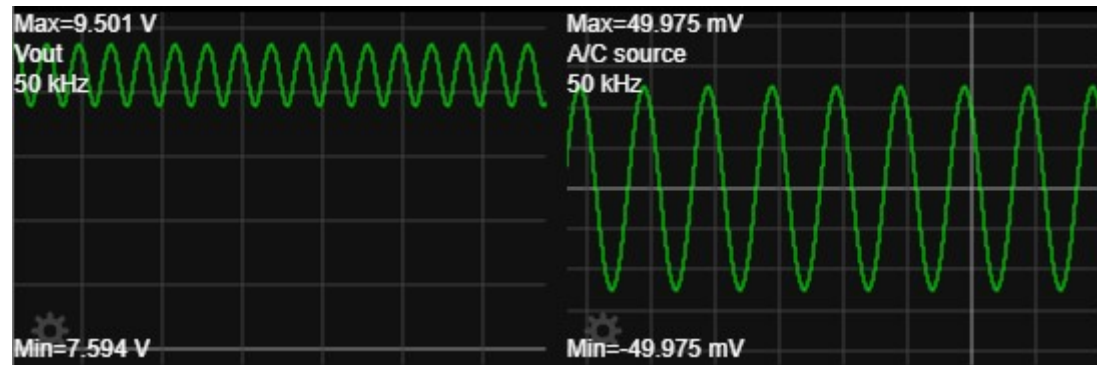


Figure 11 - 50kHz

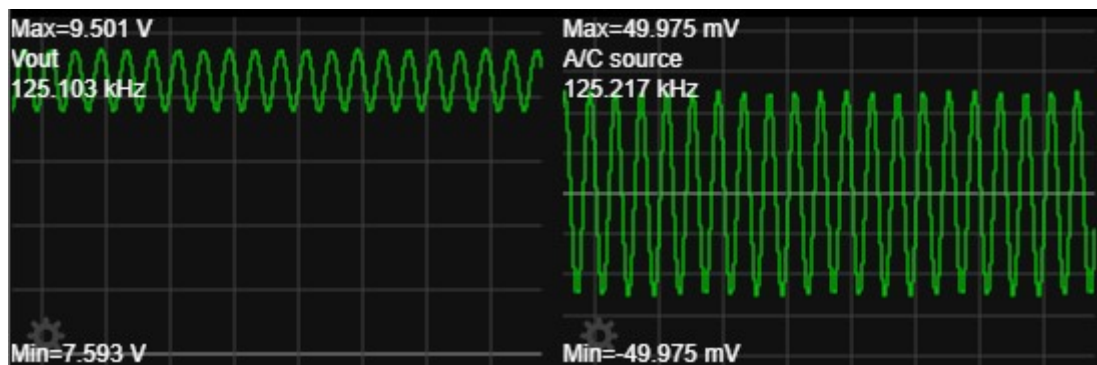


Figure 12 - 125kHz

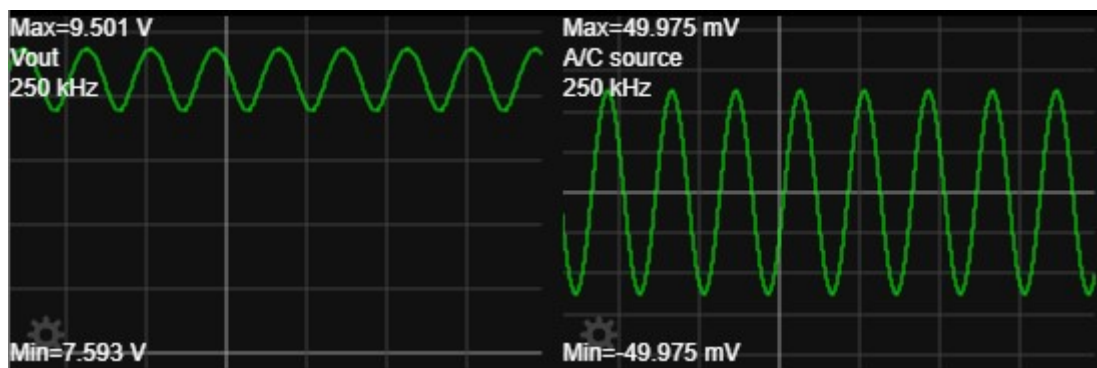


Figure 13 - 250kHz

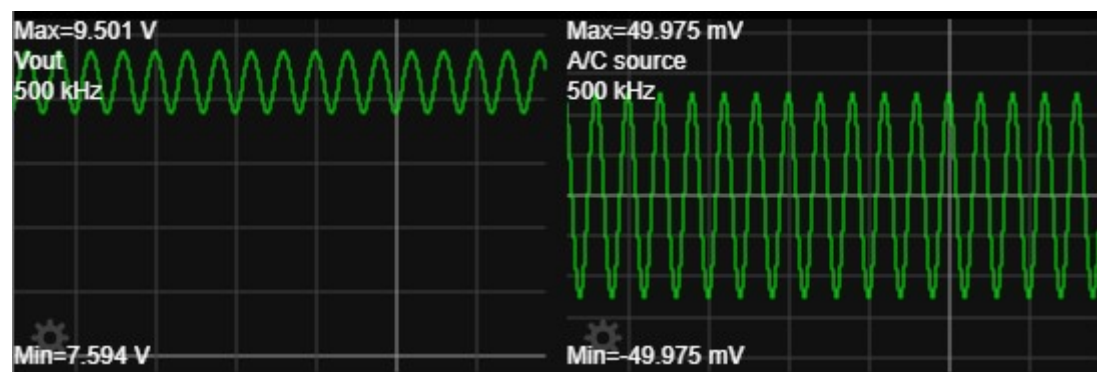


Figure 14 - 500kHz

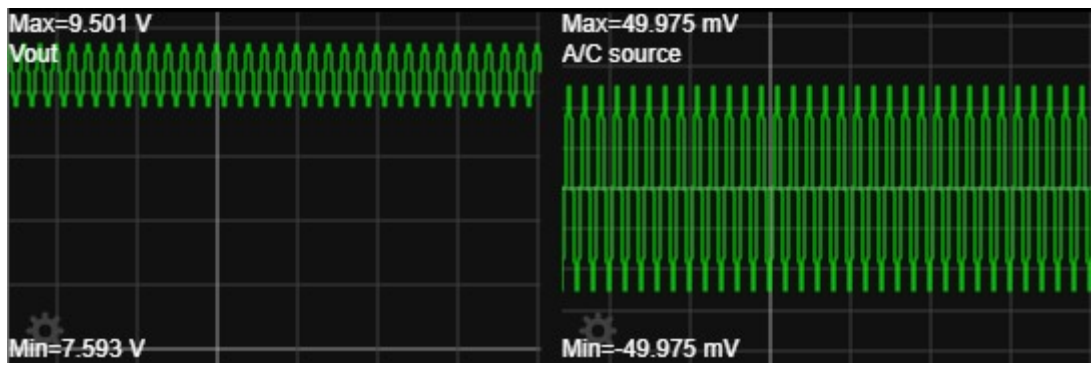


Figure 15 - 1MHz

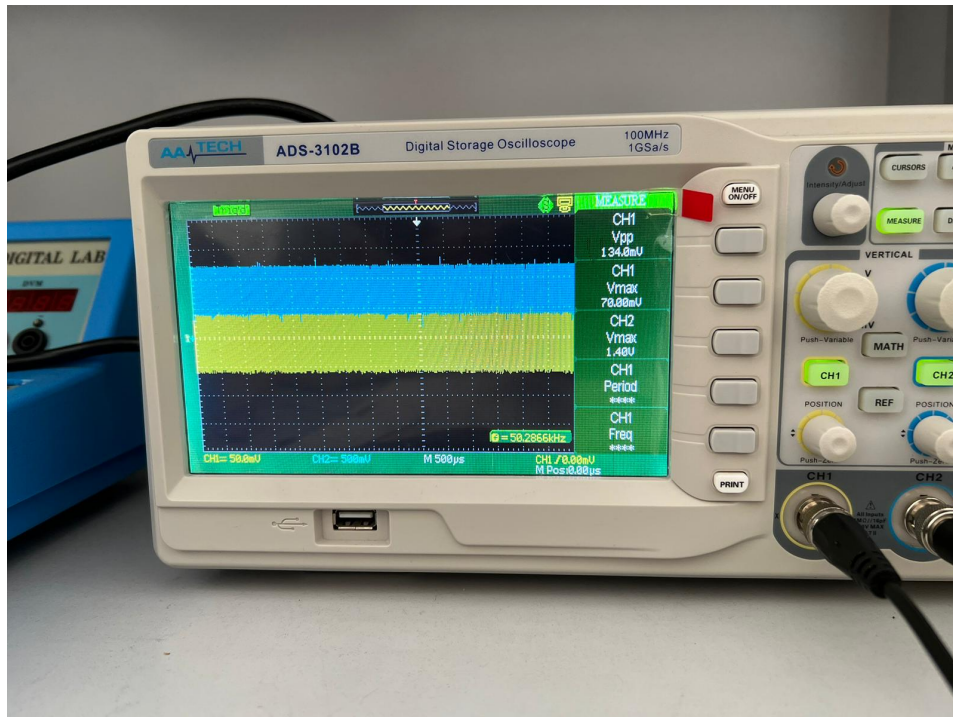


Figure 16

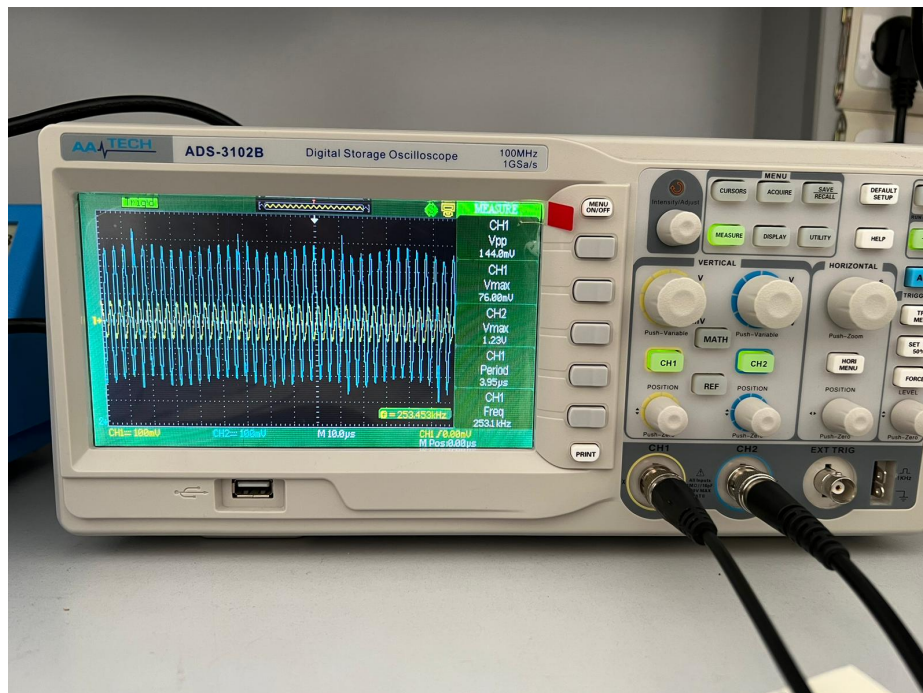


Figure 17

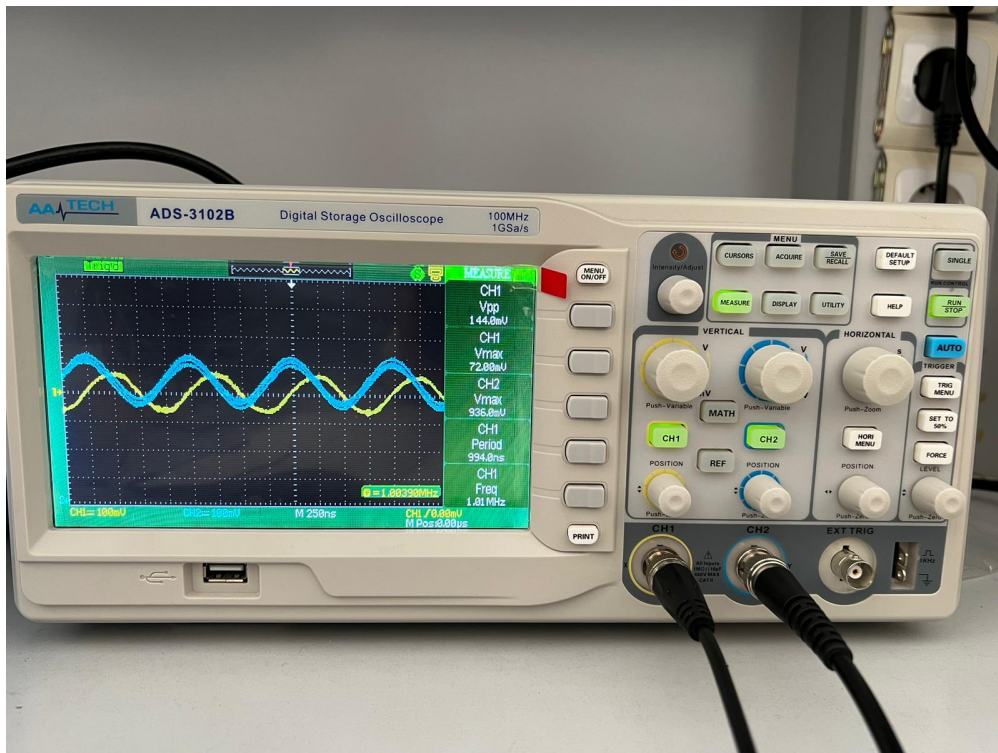


Figure 18

Unfortunately, we couldn't add all the frequency levels on the oscilloscope since there was something wrong with our equipment from the test we made as a group and as the final decision of Mr. Ahmet.

Conclusion

In this experiment, we learned the working principals of BJTs when sinusoidal voltage is applied across them. Most of our oscilloscope footage was incorrect due to reasons we and Mr. Ahmet couldn't identify clearly. It was probably due to faulty equipment.